YE Futao

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EDUCATION

The Hong Kong Polytechnic University Fall 2026 - Present
Master of Science in Metaverse Technology (Expected graduation: Aug-Oct 2027)

South China Normal University Sep 2022 - Sep 2026
Bachelor of Science in Software Engineering
- GPA: 3.41 / 5.0 | Average: 84.1 / 100

Publications
- Ye, F. (2025). Rewriting K-Means Clustering Algorithm Using DataFrames API. ICCSM 2025, 457–460.
- Ye, F. (2025). A Survey of Research Advances in Federated Learning for Anomaly Detection. Science and Technology of Engineering, Chemistry and Environmental Protection, 3(5).

PROJECTS

Robotic Car Assembly and Visual Navigation
- Assembled a robotic car for line following and obstacle manoeuvring; used Python/OpenCV for visual guidance and colour-signal detection, with HiwonderSDK motor control.

License Plate Recognition System with Python and OpenCV
Course Project
- Built a pipeline for mainland Chinese and Hong Kong plates using HSV masking, morphology, perspective rectification and projection-based character segmentation.
- Combined image difference, edge similarity and contour matching for template recognition; visualized intermediate outputs for debugging.

Road Assistance System for Visually Impaired Users Mar 2024 - Jun 2025
Innovation and Entrepreneurship Project
- Prepared and augmented segmentation masks from two datasets; trained and evaluated DeepLabv3+ for road-scene segmentation.
- Built an Android location and route-guidance prototype with interfaces reserved for future model integration.

Movie Recommendation System
Recommendation Algorithm and Backend Development
- Applied collaborative filtering and developed Java backend features with Spring Boot and MyBatis, including movie queries, filtering and user favorites.

Synchronous SGD Simulation and Optimization Experiments May 2025 - Dec 2025
Online Research Programme
- Simulated multiple SGD workers on one machine using NumPy and scikit-learn; compared optimization methods and examined hyperparameter effects on convergence.

INTERNSHIPS

Nanhua Instrument Co., Ltd. Sep 2025 - Dec 2025
IoT Software Development Intern
- Supported device monitoring and data visualization for smart gas cabinets; integrated EMQX/MQTT communication between devices and cloud services.

Foshan Electric Power Design Institute Jan 2025 - Feb 2025
Research and Development Intern
- Supported requirements analysis, first-version design and technical validation of an internal innovation competition platform.

ADDITIONAL INFORMATION
- Programming and tools: Python, Java, OpenCV, NumPy, scikit-learn, Spark DataFrame API, Spring Boot, MyBatis.
- Practice: Image processing, semantic segmentation, collaborative filtering, clustering and regression.
- Certification: Huawei Advanced HarmonyOS Developer (Oct 2024). | Cantonese: Proficient.